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MAJOR LEAGUE HACKING
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HACKTOBERFEST 26
Team Name: HalwaCoded
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Problem Statement: AI Powered
Voice Guided Form Assistant

PROPOSED SOLUTION

The Problem

- Complex, formal language makes standard forms difficult to understand.
- Language barriers and technical jargon cause confusion and delays.
- Unclear field requirements lead to errors or incomplete submissions.

Our Solution: FormSaathi

- AI assistant that guides users through forms step-by-step, one question at a time.
- Explains questions in simple language and supports voice or text responses.
- Automatically maps answers to fields with a full user review before submission.

USER FLOW / WORKING

01. Select a Form
The user uploads or opens a supported form.

02. Understand
AI explains current questions in simple, accessible language.

03. Respond
The user answers naturally through voice or text.

04. Fill
FormSaathi maps the response to the appropriate field.

05. Validate
Identifies missing fields and basic inconsistencies.

06. Review
The user checks the completed form before finishing.

Demo Flow Scenario

Open sample form → Explain question in Hindi → Spoken answer → Fill field → Flag missing field → Display reviewed form

TECHNICAL APPROACH

System Architecture Flow

PDF / Scanned Form → Extraction + OCR → AI Processing → Voice / Text UI → Field Mapping → Validation → Review Screen

Frontend & Backend

- Next.js with TypeScript & Tailwind CSS
- Lightweight API routes to coordinate model calls & form processing

AI & Voice Engine

- Gemma AI model (subject to API access & free-tier limits)
- ElevenLabs API for interactive voice response

Form & Validation

- PDF/text extraction & OCR for scanned documents
- Automated field mapping & rule-based consistency checks

AI-Assisted Development: Leveraged AI coding agents and development tools to accelerate implementation, debugging, and integration.
Fallback Strategy (Demo Mode): If Gemma or ElevenLabs APIs are restricted, predefined explanations, sample responses, and simulated voice mode are used to demonstrate core workflow.

INNOVATION & DIFFERENTIATION

Conversational Filling
Guides users through one question at a time.

OCR Processing
Supports scanned documents & text forms.

Voice & Text UI
Multiple ways for users to provide answers.

Plain Language
Makes complex questions easy to understand.

Context-Aware Assistance
Structures responses by requested info.

Validation Before Finish
Highlights missing fields & inconsistencies.

Human-Controlled Flow
Keeps answers editable & under user control.

Key Differentiator
FormSaathi combines document understanding, OCR, conversational assistance, form filling, and validation into a single guided workflow.

FEASIBILITY, IMPACT & FUTURE

Feasibility

- Controlled sample form & limited fields
- Text extraction & OCR for selected types
- Integrated AI & voice where feasible
- Fallback mode for API limits/outages
- Accelerated via AI coding agents
- Prioritizes reliable, full demonstration

Expected Impact

- Reduces confusion around complex forms
- More approachable for language barriers
- Helps identify missing info early
- Foundation for accessible digital services

Future Scope

- Expand Indian-language support
- Improve OCR & additional document types
- Integrations with public-service workflows
- Strengthen privacy & validation features

Vision: Make complex forms easier for everyone, one question at a time.

Implementation Priority: Build full happy path first (one form, OCR, simple explanation, voice/text answer, field population, validation). Integrate Gemma/ElevenLabs with fallback ready.